# ATOR DFIR - Incident Investigation Report

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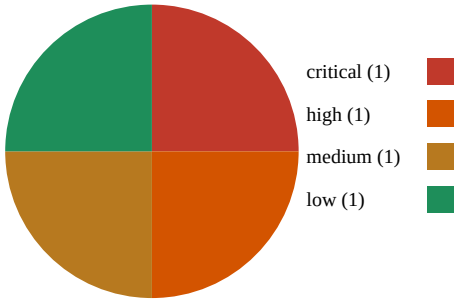
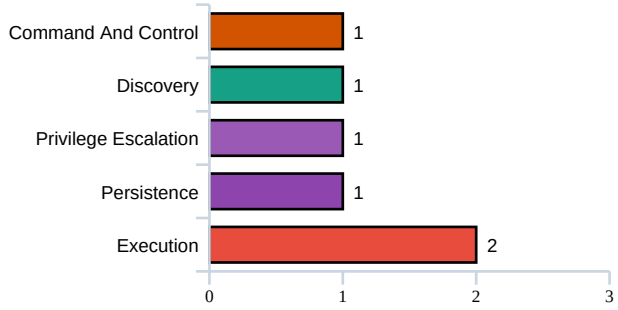
| | |
|---|---|
| Verdict | **CONFIRMED COMPROMISE INDICATORS - MULTI-STAGE ATTACK CHAIN** |
| Risk Level | CRITICAL |
| Host | ator-win10-host (windows) |
| Agent Version | 1.0.0-e2e |
| Attack Stages Observed | 5 |
| Detections (C/H/M/L) | 1 / 1 / 1 / 1 |
| Generated (UTC) | 2026-10-20 02:25:14 |

## Executive Summary

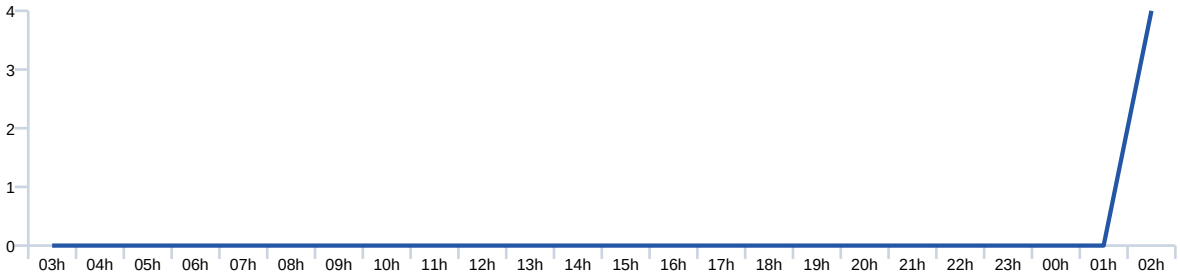
An automated investigation of endpoint **ator-win10-host** identified 4 detection(s) across 5 ATT&CK tactic stage(s). The overall assessment is **CRITICAL**: CONFIRMED COMPROMISE INDICATORS - MULTI-STAGE ATTACK CHAIN. Recommended actions: preserve evidence manifests (section 7), review the attack chain (section 3) and technical annex (sections 5-6), and apply containment through the approval workflow if compromise indicators are confirmed. All timestamps are normalized to UTC.

## Detection Analytics




Detections per hour - last 24 hours (UTC):




## Observed Attack Chain (Source of Compromise)

| # | Tactic | Techniques (click ID for MITRE reference) | Observed Window (UTC) |
|---|---|---|---|
| 1 | Execution | T1059.001 PowerShell (x1)<br>T1053.005 Scheduled Task (x1) | 2026-10-20 02:09:30<br>→ 2026-10-20 02:09:30 |
| 2 | Persistence | T1053.005 Scheduled Task (x1) | 2026-10-20 02:09:30<br>→ 2026-10-20 02:09:30 |
| 3 | Privilege Escalation | T1053.005 Scheduled Task (x1) | 2026-10-20 02:09:30<br>→ 2026-10-20 02:09:30 |
| 4 | Discovery | T1033 System Owner/User Discovery (x1) | 2026-10-20 02:09:30<br>→ 2026-10-20 02:09:30 |
| 5 | Command And Control | T1071.001 Web Protocols (x1) | 2026-10-20 02:09:30<br>→ 2026-10-20 02:09:30 |

## ATT&CK Coverage Matrix (observed tactics highlighted)

| | | | | | | |
|---|---|---|---|---|---|---|
| Reconnaissance | Resource Development | Initial Access | **Execution *** | **Persistence *** | **Privilege Escalation *** | Defense Evasion |
| Credential Access | **Discovery *** | Lateral Movement | Collection | **Command And Control *** | Exfiltration | Impact |

* = tactic observed on this host

## Root Cause Analysis (What / Where / When)

Every detection below is stated as: what fired, its inferred root cause, where it happened (host and the exact process / file / network locus), and when (first seen - last seen, UTC). Root cause is an evidence-based inference, not confirmed attribution.

| When (UTC, first - last) | What (rule) | Sev | Root cause | Where |
|---|---|---|---|---|
| 2026-10-20 02:09:30 | **Account Discovery via Whoami** | low | Host / account discovery activity | ator-win10-host (windows) process C:\Windows\System32\whoami.exe (pid 4101) |
| 2026-10-20 02:09:30 | **Scheduled Task Creation via Schtasks** | medium | Persistence mechanism registered to survive logoff/reboot | ator-win10-host (windows) process C:\Windows\System32\schtasks.exe (pid 4102) |
| 2026-10-20 02:09:30 | **Suspicious Reverse Shell Destination Ports** | critical | Outbound connection to a reverse-shell / C2 destination | ator-win10-host (windows) process powershell.exe (pid 4100); network 198.51.100.23:4444 |
| 2026-10-20 02:09:30 | **PowerShell Encoded Command Execution** | high | Obfuscated / encoded command execution | ator-win10-host (windows) process C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (pid 4100) |

## Technical Annex - Key Detections

| Time (UTC) | Type | Rule | Sev | MITRE | Detail |
|---|---|---|---|---|---|
| 2026-10-20 02:09:30 | sigma | Account Discovery via Whoami | low | T1033 | pid=4101; ppid=4100; name=whoami.exe; cmdline=whoami.exe /priv; exe_path=C:\Windows\System32\whoami.exe |
| 2026-10-20 02:09:30 | sigma | Scheduled Task Creation via Schtasks | medium | T1053.005 | pid=4102; ppid=4100; name=schtasks.exe; cmdline=schtasks.exe /create /tn AtorPersist /tr cmd.exe /sc once; exe_path=C:\Windows\System32\schtasks.exe |
| 2026-10-20 02:09:30 | sigma | Suspicious Reverse Shell Destination Ports | critical | T1071.001 | pid=4100; remote_ip=198.51.100.23; remote_port=4444; local_ip=10.0.0.2; local_port=51000 |
| 2026-10-20 02:09:30 | sigma | PowerShell Encoded Command Execution | high | T1059.001 | pid=4100; ppid=500; name=powershell.exe; cmdline=powershell.exe -nop -w hidden -enc SQBFAFgAKQAgAA==; exe_path=C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe |

## Source of Compromise Assessment

No source-of-compromise hypothesis was attached to the available detections. Review the timeline and raw evidence for delivery and user-action telemetry.

## Timeline Highlights

| Time (UTC) | Kind | Event | Detail |
|---|---|---|---|
| 2026-10-20 02:09:30 | detection | [SIGMA] Account Discovery via Whoami | {"technique_id": "T1033", "summary": {"pid": "4101", "ppid": "4100", "name": "whoami.exe", "cmdline": "whoami.exe /priv", "exe_path": "C:\\Windows\\System32\\whoami.exe"}} |
| 2026-10-20 02:09:30 | detection | [SIGMA] Scheduled Task Creation via Schtasks | {"technique_id": "T1053.005", "summary": {"pid": "4102", "ppid": "4100", "name": "schtasks.exe", "cmdline": "schtasks.exe /create /tn AtorPersist /tr cmd.exe /sc once", "exe_path": |
| 2026-10-20 02:09:30 | detection | [SIGMA] Suspicious Reverse Shell Destination Ports | {"technique_id": "T1071.001", "summary": {"pid": "4100", "remote_ip": "198.51.100.23", "remote_port": "4444", "local_ip": "10.0.0.2", "local_port": "51000", "proto": "tcp", "proces |
| 2026-10-20 02:09:30 | detection | [SIGMA] PowerShell Encoded Command Execution | {"technique_id": "T1059.001", "summary": {"pid": "4100", "ppid": "500", "name": "powershell.exe", "cmdline": "powershell.exe -nop -w hidden -enc SQBFAFgAKQAgAA==", "exe_path": "C:\ |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMSP | System32\drivers\vmswitch.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMSNPXYMP | \SystemRoot\System32\drivers\VmsProxyHNic.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmsmp | \SystemRoot\System32\drivers\vmswitch.sys |

| Time (UTC) | Kind | Event | Detail |
|---|---|---|---|
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMnetuserif | \SystemRoot\system32\DRIVERS\vmnetuserif.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMnetDHCP | C:\Windows\SysWOW64\vmnetdhcp.exe |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMnetBridge | \SystemRoot\system32\DRIVERS\vmnetbridge.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMnetAdapter | \SystemRoot\System32\drivers\vmnetadapter.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmms | %systemroot%\system32\vmms.exe |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicvss | %systemroot%\system32\svchost.exe -k LocalSystemNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicvmsession | %systemroot%\system32\svchost.exe -k LocalSystemNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmictimesync | %systemroot%\system32\svchost.exe -k LocalServiceNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicshutdown | %systemroot%\system32\svchost.exe -k LocalSystemNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicrdv | %systemroot%\system32\svchost.exe -k ICService -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmickvpexchange | %systemroot%\system32\svchost.exe -k LocalSystemNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicheartbeat | %systemroot%\system32\svchost.exe -k ICService -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmicguestinterface | %systemroot%\system32\svchost.exe -k LocalSystemNetworkRestricted -p |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmgid | \SystemRoot\System32\drivers\vmgid.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmcompute | %systemroot%\system32\vmcompute.exe |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] vmbusr | \SystemRoot\System32\drivers\vmbusr.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMBusHID | \SystemRoot\System32\drivers\VMBusHID.sys |
| 2026-10-20 02:09:30 | persistence | [PERSISTENCE service] VMAuthdService | "C:\Program Files (x86)\VMware\VMware Workstation\vmware-authd.exe" |

## Evidence Integrity (Manifests)

| Collection ID | Started (UTC) | Agent Ver | Artifacts | Manifest SHA-256 |
|---|---|---|---|---|
| f991d3ac-c496-414b-8943-afd403f89bf8 | 2026-10-20 02:07:18 | 1.0.0 | 1342 | 48c49312a5cdc5bc9140260bcc59d30c9772bc5274363f6709d6a58a273611d1 |
| win-atk-001 | 2026-10-20 02:09:30 | 1.0.0-e2e | 6 | b32675b6755be2a2339c63e58dd09e50587b16b29392bc894662af187aa25f6c |

Timeline events analyzed: 699; time-skew anomalies flagged: 0. Every manifest SHA-256 is verified server-side at ingestion; hashes above let recipients re-verify collection integrity independently.

## Deep-Dive Artifacts (Velociraptor)

No Velociraptor artifacts were collected from this endpoint. Sections above therefore rest on routine agent telemetry alone - absence of a finding here means the deep-dive artifacts were never run, not that they came back clean.